

Embedded Systems Report

Prepared using the uploaded source documents on Embedded Systems, Gesture Vocalizer Project, and Arduino Source Code.

This report provides an overview of embedded systems and presents a mini-project called "Gesture Vocalizer for Deaf and Mute People using Arduino." The report also includes the working principle, components, advantages, and sample Arduino source code.

1. Introduction to Embedded Systems

Embedded systems are specialized computing systems designed to perform a specific function or set of functions within a larger system. Unlike general-purpose computers, embedded systems are dedicated, efficient, and often operate in real time. An embedded system generally includes: Microcontroller or Microprocessor Memory Input and Output Interfaces Embedded Software or Firmware Examples of embedded systems include washing machines, microwave ovens, automotive control units, medical devices, smart televisions, and industrial automation systems.

Types of Embedded Systems Standalone Embedded Systems Real-Time Embedded Systems Networked Embedded Systems Mobile Embedded Systems **Key Characteristics** Dedicated functionality Low power consumption Compact size High reliability Real-time operation

2. Gesture Vocalizer Project

The Gesture Vocalizer project is designed to help deaf and mute individuals communicate more effectively with people unfamiliar with sign language. The system uses flex sensors attached to a glove to detect finger movements and hand gestures. The sensor values are processed by an Arduino Uno microcontroller, which identifies predefined gestures and converts them into text or voice messages. The generated message can be displayed on an LCD screen and transmitted using Bluetooth or speakers.

Problem Statement

Many deaf and mute individuals rely on sign language for communication. Since most people do not understand sign language, communication becomes difficult in hospitals, schools, and public places. This project helps bridge the communication gap by converting gestures into understandable outputs.

Working Principle Flex sensors detect finger bending. The resistance change is converted into voltage signals. Arduino reads and processes sensor values. The system compares values with predefined gestures. The corresponding message is displayed or spoken.

Component	Purpose
Arduino Uno	Main controller for processing sensor data
Flex Sensors	Detect finger bending and gestures
16x2 LCD Display	Displays output messages
HC-05 Bluetooth Module	Wireless communication
Resistors	Used in voltage divider circuit
Battery	Power supply
Glove	Mounting platform for sensors

Estimated Budget: ■2500 – ■3000

Advantages Helps deaf and mute people communicate easily. Low cost and easy to implement. Portable and user-friendly. Can be extended using AI-based gesture recognition.

3. Arduino Source Code

The following Arduino program displays predefined gesture messages on a 16x2 LCD display using an I2C interface.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

const char* messages[] = {
  "WASHROOM",
  "DRINK WATER",
  "FOOD",
  "FRUIT",
  "THANK YOU",
  "HELP",
  "SORRY",
  "YES",
  "NO",
  "PLEASE",
  "WAIT",
  "GOOD MORNING"
};

void setup() {
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
  lcd.print("SIGN LANGUAGE");
  delay(2000);
  lcd.clear();
}

void loop() {
  // Display messages on LCD
}
```

Conclusion

Embedded systems play an important role in modern technology and automation. The Gesture Vocalizer project demonstrates how embedded systems can be used to solve real-world communication challenges. By integrating sensors, microcontrollers, and display modules, the system provides an affordable and practical communication solution for deaf and mute individuals.